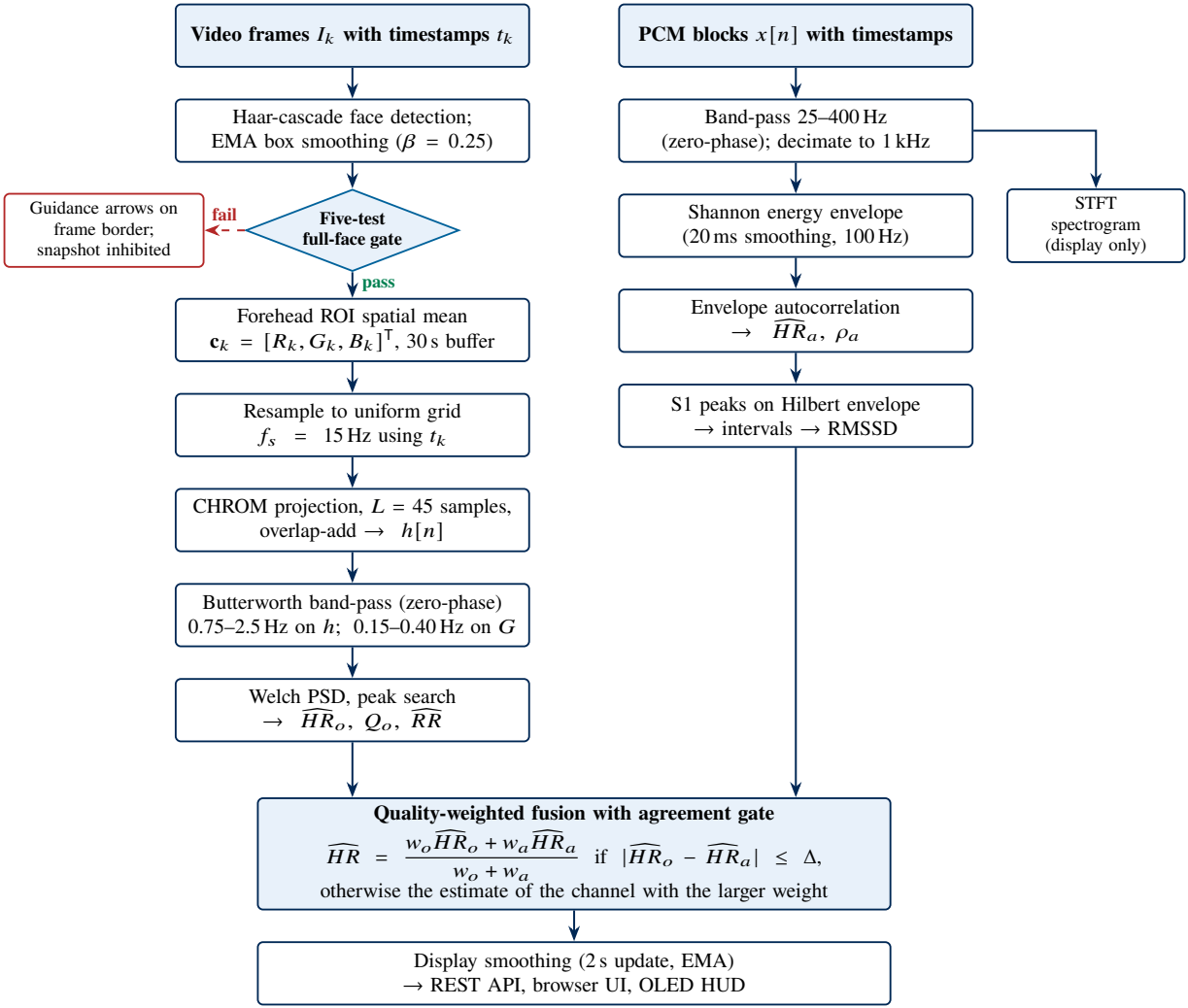

Video frames I_k with timestamps t_k
Haar-cascade face detection; EMA box smoothing ($\beta = 0.25$)
Five-test full-face gate
fail
Guidance arrows on frame border; snapshot inhibited
pass
Forehead ROI spatial mean $\mathbf{c}_k = [R_k, G_k, B_k]^\mathsf{T}$, 30 s buffer
Resample to uniform grid $f_s = 15$ Hz using t_k
CHROM projection, $L = 45$ samples, overlap-add $\rightarrow$ $h[n]$
Butterworth band-pass (zero-phase) 0.75–2.5 Hz on h; 0.15–0.40 Hz on G
Welch PSD, peak search $\rightarrow$ $\widehat{HR}_o$, Q_o, $\widehat{RR}$
PCM blocks $x[n]$ with timestamps
Band-pass 25–400 Hz (zero-phase); decimate to 1 kHz
STFT spectrogram (display only)
Shannon energy envelope (20 ms smoothing, 100 Hz)
Envelope autocorrelation $\rightarrow$ $\widehat{HR}_a$, ρ_a
S1 peaks on Hilbert envelope $\rightarrow$ intervals $\rightarrow$ RMSSD
Quality-weighted fusion with agreement gate
$\widehat{HR} = \frac{w_o \widehat{HR}_o + w_a \widehat{HR}_a}{w_o + w_a}$ if $|\widehat{HR}_o - \widehat{HR}_a| \le \Delta$,
otherwise the estimate of the channel with the larger weight
Display smoothing (2 s update, EMA)
$\rightarrow$ REST API, browser UI, OLED HUD